Queensland's Million Wheel Battery

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1. Executive Narrative

Queensland has a 525MW solar sucking battery that moves and grows each day. It is divided into 7kW clusters, each with a set of four wheels, motor, and private driver for maximum flexibility in reassembling to the electricity grid's requirements each day. Unfortunately this battery hasn't been assembled out of the entropy soup yet. It could be for as little as $375m – competitive on cost, capability and value against other recent grid-scale battery projects across Queensland.

Every morning, the state's 75,000 electric vehicles[1] set out on their daily routines. Many of them will be parked while their owners are at work, right through the midday solar peak. An inexpensive, managed, public network of 7kW AC chargers would match the needs of grid operators (providing a reasonable estimate of 87.5MW of distributed, rapidly controllable solar-soak load) and drivers (who maintain free agency but can be trivially herded around the electricity network with pricing discounts). This can be connected incrementally at approximately $1m per MW of distributed, controllable load.

This unique load offers two levels of demand management control: at a high level, AC charge outlets can be installed intentionally across the network of substations and feeders. But by offering dynamic pricing (much like petrol stations) and managed charging (control over the rate and schedule of charge in exchange for discounted prices), grid operators can nudge drivers to adjust where and when that demand arrives on a daily basis. High utilisation of this equipment is the single metric that completely aligns the incentives of both drivers and grid operators.

As EV adoption expands, managing charge demand becomes an increasingly urgent priority. Today's 87.5MW opportunity is based on a statewide fleet that is only 1.6% electrified. The Ergon/Energex 2025 Strategic Forecast anticipates the EV fleet growing to 1.66m (base case) within 10 years[2] with proportional growth in reasonably anticipated charging load to 2GW. At this scale, the Million Wheel Battery becomes a massive distributed shock absorber for everything else, and transforms the electricity grid from a high-frequency trading game into a proactively scheduled milk run.

The purpose of this report is to argue the case for assembling the "Shape Shifting Statewide Solar Soak Battery With A Million Wheels" and outline an initial pilot implementation using DMIA funds.

2. Background (why now is the time)

Demand management and EV charging have historically been separate concerns. A confluence of factors over the last year has made this proposal not only viable, but unavoidable.

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[1] **Queensland Government** (Electric vehicle snapshot December 2025)
https://www.qld.gov.au/transport/projects/electricvehicles/electric-vehicle-snapshot/electric-vehicle-snapshot-december-2025

[2] **Ergon | Energex** (2025 Strategic Forecast Annual Report)
https://www.energex.com.au/__data/assets/pdf_file/0019/1150246/Strategic-Forecasting-Annual-Report-2025.pdf

2.1 Solar Surplus

Ergon's own planning reports document a 40% decline in network minimum demand over three years (862MW in 2022 to 518MW in 2025)[3], driven by daytime solar electricity production. For Energex, it's even more stark: a 90% drop from 399MW to 40MW over the same period[4]. The Strategic Forecast anticipates negative minimum demand for Energex this year, and for Ergon around 2033/2034. Voltage rise and feeder congestion in solar-dense areas are already here. The federal Solar Sharer Offer independently acknowledges the same structural need: shifting evening demand toward the midday solar peak.

2.2 EV Charging

Public EV charging policy has traditionally chased range anxiety with expensive DC fast chargers on highway corridors. While DC chargers are crucial, the primary mode of charging that is being enabled is AC charging, at home[5]. The National Electric Vehicle Strategy[6] imagines AC charging as a residential activity covering 80% of total charging; the Electric Vehicle Council's Home Charging report[7] similarly frames AC charging as a residential activity as a grid impact to be mitigated. Networks have had a reactionary response that aims to move the home charging load outside the evening peak (controlled load tariffs; dynamic household AC charging connections; Queensland Electricity Connection Manual version 4).

EV policies consistently avoid the simple infrastructure barrier to fully aligning EV charging with grid interests as a massive solar soak opportunity: most vehicles lack the facility to plug in during the day. Public AC charging is imagined as a soft consumer amenity, rather than a powerful grid management opportunity.

Many households justify the expense of an AC charger in their private capacity, which requires no coordination beyond a single electrician. This same class of charging equipment should be easy enough to install where vehicles park during the day, yet this requires alignment between property owners, equipment owners, network operators, and funding bodies. Individuals who would use these facilities are ill-equipped to coordinate their provision. Without an institutional champion, public AC charging has remained a poorly-funded, uncoordinated, dissatisfying driver experience.

2.3 The combined opportunity

Queensland's EV fleet has crossed the threshold where EV charging becomes operationally meaningful as a demand management tool. Like the Federal Government and EVC, the 2025 Ergon/Energex Demand Management Plan[8] considers EV charging as primarily residential, behind-the-meter behaviour. A public daytime AC destination charging network would completely transform the electricity grid. It must be considered as both essential infrastructure and a demand management instrument, and Energy Queensland is in the ideal position to balance and pursue these mandates simultaneously.

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[3] **Ergon** (Distribution Annual Planning Reports; minimum demand figures)
https://ergon.com.au/network/about-us/company-reports,-plans-and-charters/distribution-annual-planning-report
[4] **Energex** (Distribution Annual Planning Reports; minimum demand figures)
https://www.energex.com.au/about-us/company-reports,-plans-and-charters/distribution-annual-planning-report
[5] **Ergon | Energex** (EV SmartCharge Queensland Insights Report)
https://www.energex.com.au/__data/assets/pdf_file/0008/1096496/EV-SmartCharge-Queensland-Insights-Report.pdf
[6] **Department of Climate Change, Energy, the Environment and Water** (National Electric Vehicle Strategy)
https://www.dcceew.gov.au/sites/default/files/documents/national-electric-vehicle-strategy.pdf
[7] **Electric Vehicle Council** (Home Charging and the Grid)
https://electricvehiclecouncil.com.au/wp-content/uploads/2024/12/Home-EV-Charging-and-the-Grid-2030_-edition-2.1_all-edits-complete-MS.pdf
[8] **Ergon | Energex** (2025 Demand Management Plan)
https://www.energex.com.au/__data/assets/pdf_file/0019/1085005/Demand-Management-Plan-2025-26.pdf

3. The Million Wheel Battery: Statewide Vision

3.0 EV electricity consumption and capital expenditure model

Let's start with a sensible average of daily statewide electricity consumption (7kWh per vehicle) and consider what aggregate charging behaviour could look like:

	2025	2035
Start with the average daily consumption per EV and subsequent charge time at 7kW:	7kWh[9] ⇒ 1 hour at 7kW	
In order to stretch the charge time over a 6-hour window (the 9am-3pm solar peak), let's have each car plug in once every 6 days, but charge 6x as much. This makes the charge session:	42kWh ⇒ 6 hours at 7kW	
The total state EV fleet is	75,000 cars[10]	1,660,000 cars[11]
…so ⅙ of the fleet charging on a given day is	12,500 cars	276,666 cars
…(for reference, the statewide daily EV electricity consumption is)	525MWh	11.6GWh
The aggregate statewide load, at 7kW per vehicle, is	87.5MW	2GW
Let's provision 1.5x more chargers than charging vehicles (see: 3.4, Utilisation Metric)	18,750 chargers	420,000 chargers
…at $5000 per installed charger[12], total hardware capex is:	$94m	$2.1b
In both cases, the facility to attract aggregate load costs about $1m per MW.		

The 2025 Strategic Forecast anticipates a mere 250MW of EV charging offset to minimum demand by 2035. The difference between that and the 2GW opportunity presented here is probably due to different assumptions: the former assumes the limits of home-based charging and a demand strategy that only aims to shift load away from the evening peak. The latter recognises the full potential that can be unlocked with an infrastructure rollout.

Note that the cost/outcome numbers are parameters that can be tweaked in this model – charger buildout scales linearly with both plug-in interval, aggregate power draw (demand enabled), and capex. An alternative 2035 configuration could look like a 3hr/21kWh charge window every 3 days, pulling 5.8GW across 830,000 charge plugs, costing $4.15b.

Further note that the cost to implement dynamic pricing and managed charging (which we explore later) will depend on existing capabilities and can be scoped progressively as sophistication is built out over time.

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[9] 5-6km per 1kWh (Queensland Government Electric Vehicle Snapshot June 2025); 33km Queensland daily average (Australian Bureau of Statistics, Survey of Motor Vehicle Use 2021)
[10] **Queensland Government** (Electric vehicle snapshot December 2025)
[11] **Ergon | Energex** (2025 Strategic Forecast Annual Report), "medium" growth scenario
[12] **7kW charging unit** – $2000 for commodity hardware, $3000 for labour and capital works
https://wholesale.jetcharge.com.au/products/wallbox-pulsar-pro-7kw-tethered-7m

3.1 Economic comparison with battery projects and augmentation deferral

The essential claim of 3.0 is that Energy Queensland can build infrastructure that induces distributed midday demand granularly at a capital cost of $1m per MW. At this rate, the Million Wheel Battery is highly competitive against recent small- and large- scale battery projects by Energy Queensland:

Project	Cost	MW / MWh	Owner/Delivery	Location	Wheels
Million Wheel Battery, 2025	**$94m**	**87.5MW / 525MWh** *charge only "sink"; expected real-world load*	**Yurika**	**Distributed, with a focus on solar-dense sections of grid**	**1,000,000**
Local Network Battery Program Stages 1 & 2[13]	~$150m	68MW 136MWh	Ergon/Energex/ Various	Bundaberg, Hervey Bay, Toowoomba ×2, Townsville ×2, Yeppoon, Cairns ×2, Mackay, Emerald, Mundubbera, Gladstone, Howard, Raby Bay, Morayfield, Bribie Island	0
Local Network Battery Program Stages 3 & 4[14]	$179m	48MW 96MWh + ~4MWh flow	Ergon/Energex/ Various	Mooloolaba, Runaway Bay, Cornubia, Jimboomba, Woodridge, Yatala, Barcaldine, Dalby, Glenella, Toowoomba, Milchester, Maryborough + Burrum Heads, Ipswich (flow)	0
Subtotal (LNBP)	*~$329m*	*116MW / 236MWh*			*0*
Swanbank BESS[15]	$330m	250MW / 500MWh	CleanCo/Yurika	Ipswich	0
Tarong BESS[16]	$514m	300MW / 600MWh	Stanwell/Yurika	Nanango	0
Stanwell BESS[17]	$747m	300MW / 1,200MWh	Stanwell/Yurika	Rockhampton	0
Subtotal (BESS)	*$1.59bn*	*850MW / 2,300MWh*			*0*
Total (LNBP+BESS)	*~$1.92bn*	*~966MW / ~2,536MWh*			*0*

It has unique structural advantages too: battery asset ownership risk, depreciation, and maintenance liability rests entirely with private individuals, not the network operator. Because the electricity flow is one-way only, this avoids the cost and complexity of export-capable inverters, protection relays, or bidirectional metering. The distributed placement of load across the grid is especially valuable for a sparse regional network like Ergon's, which has a proportionally high investment in sub-transmission assets[18]. Together with its personal chauffeurs, the Million Wheel Battery can even be induced to re-balance across the feeder networks at the whims of the operator in a way that hot water tanks cannot.

Strategically, the Million Wheel Battery drives augmentation deferral on both ends of the demand spectrum: it reduces the need to augment residential networks struggling to deliver evening EV charge load, while simultaneously reducing the need to augment to accommodate midday solar saturation. For the sake of comparison, the 2025-2030 approved augmentation budget is $432.9m for Ergon and $538.6m for Energex[19] .

[13] Estimated. Specific program cost unknown.
[14] https://statements.qld.gov.au/statements/99503
[15] https://statements.qld.gov.au/statements/99735
[16] https://www.stanwell.com/tarong-battery
[17] https://www.stanwell.com/stanwell-battery
[18] **Ergon | Energex** (2025 Strategic Forecast Annual Report), link above
[19] **Australian Energy Regulator** (Ergon, Energex DMIA/capex budget approval)
www.aer.gov.au/industry/registers/determinations/ergon-energy-determination-2025-30
https://www.aer.gov.au/industry/registers/determinations/energex-determination-2025-30

3.2 Demand Management Across Space and Time

The Million Wheel Battery offers network operators two distinct spatial flavours of control that fixed assets cannot provide. At an infrastructure level, network operators can choose where to install AC chargers. They can be distributed across substations and feeder networks that most urgently need to manage localised daytime solar production. The dynamic operational version of this comes when the AC charger network is dense enough that price signals can be used to nudge drivers to connect to alternative points on the electricity network according to daily operational needs.

Temporal control begins at a coarse level. Price signals can persuade drivers to plug in during specific windows (eg. 10am-2pm) or on days with high forecast solar production. For higher-resolution control, AC charging hardware that complies with the OCPP 2.1 open standard allows grid operators to send charging profiles (power delivery levels over time) to individual charge units. These profiles can be tailored to both live feeder conditions and the attached vehicle's state-of-charge (capacity to accept power).

This is a qualitatively different capability from Ergon's existing toolkit. Economy tariffs, Tariff 31/33, and the Residential Scheduled EV Charging Tariff are time-scheduled instruments with no real-time modulation capacity or spatial resolution.

3.3 The Driver Proposition

Let's view this AC charge network from the driver's seat.

A dense network of AC chargers means drivers can plug in during a range of high-dwell daytime activities. These include workplace carparks; shopping centers; leisure and recreation places like popular swimming holes. The fact that these options have been intentionally distributed across the feeder network is invisible to drivers.

Many drivers have a great deal of flexibility to shift their charge: the previously assumed 7kWh per day is under 20% of a 40kWh vehicle battery on the low end (long range Tesla variants push about 80kWh). They can choose to charge at the swimming pool or the shopping center, or on a sunny Thursday rather than a rainy Wednesday. Commuters who plug in at 9am suffer no inconvenience if their charge is deferred until noon and complete by 5pm. Modulating a 7kW charge session by 10% would be imperceptible to individual drivers but could benefit the grid in aggregate. Drivers who can't charge at home (renters or apartment dwellers) can now incorporate public charging into their daily routine. Drivers who need full-power, immediate charge delivery are accommodated too: the billing interface should present them with a full-power option at a standard, non-discounted rate.

It's no secret that energy consumers love a good deal. Fossil fuel drivers use apps like PetrolSpy and Fuel Map to spot the cheapest service stations. They also play games with discount dockets where a standard 4 cent per litre (~2%) discount sways millions of purchase decisions daily. The UQ Charge-EV study[20] found that consumers shifted the timing of their at-home charging to off-peak times in response to a 20c/kWh incentive; a public AC charge network would allow them to shift both time and location.

The Million Wheel Battery only works if drivers use it. Luckily, their flexibility and equipment are both valuable contributions to the demand management game which are rewarded with pricing that's preferential or possibly even free (consistent with the Federal Solar Sharer Offer[21]). Because this demand management capacity is now critical to grid stability and backed by SLA, AC charging equipment faults are

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[20] **La Nauze, A.,** Philip, T., Lim, K., Friesen, L., Menezes, F., & Page, L. (2024), UQ CHARGE-EV Project Final Report. The University of Queensland, Australia https://aibe.uq.edu.au/files/10275/UQ-CHARGE-EV-Project-Final-Report.pdf

[21] **Australian Government** (Solar Sharer Offer) minister.dcceew.gov.au/bowen/media-releases/next-steps-solar-sharer-offer

treated with the same urgency as a traditional network failure. Drivers can reasonably expect to rely on the public electricity network just as much as residential consumers.

3.4 The Utilisation Metric

The effective capacity of the Million Wheel Battery depends on how many vehicles are plugged in, but there is the age-old tradeoff between utilisation and flexibility to be made.

A highly utilised network is a sign of mutual convenience for all parties, however, 100% utilisation means 0% availability. This is a bad outcome for drivers who want to charge, and the grid operator loses one of the two degrees of flexibility shifting load spatially. In practice, ideal utilisation will be high but with sufficient headroom to accommodate demand surges or load shifting when required. Determining an ideal utilisation range that also suits drivers will be one of the pilot's primary outputs, but the costings in 3.0 propose a 1.5x overprovisioning of chargers relative to target load.

3.5 Comparison with DC Fast Charging

The DCFC-dominated rollout has created a widespread assumption that more charging power is better.

Using the NRMA Highway program as a reference, $1m will purchase approximately 3-6 DCFC plugs over one or two sites[22]. This covers expensive high-power hardware, transformer and connection upgrades, and associated civil works. The nameplate capacity of 6x 200kW chargers might be 1.2MW, but in practice, vehicles gradually taper down their power draw as the battery fills up. The load profile is sharp and spiky: drivers arrive sporadically, demand full power immediately, and might spend 10-30 minutes charging.

That same $1m would buy 200 AC plugs which can be distributed according to existing transformer utilisation. At 100% plug utilisation, the nameplate capacity of 1.4MW is completely available to the grid operator to manage, because vehicles will happily draw the full 7kW until about 95%+ state-of-charge. The load profile is more granular, smooth and sustained compared to the DCFC scenario even without the proposed management techniques.

3.6 The Milk Run Electric Grid Model, and Other Future Considerations

As the Million Wheel Battery is built out, more sophisticated demand management strategies become possible. For instance, drivers might reserve a charging outlet for a specific time window: a strong predictive signal for grid operators of when and where to expect load. In aggregate, this makes the demand management operation look less like a reactionary high-frequency trading game, and more like a proactively scheduled milk delivery run. With a daily forecast of how much energy needs to be delivered and where, grid operators can plan that delivery in a relaxed fashion.

In the 2035 scenario, the daily delivery of 11.6GWh can be pumped through a midday envelope of several GW. During this window, the 'milk run' could be viewed as the primary task of the electricity grid. Instantaneous, inflexible demand (lights and appliances) stops being a driver of grid stress and becomes a statistical afterthought easily absorbed into the day's milk delivery schedule. Everyone is happy as long as the milk gets delivered by the agreed time.

As the Million Wheel Battery grows as a grid management asset, security and failure modes will become necessary considerations. If pricing signals are sent over the internet, operators need to consider control plane failsafe scenarios. One strategy might have isolated charging units revert to a lower power draw or

[22] **NRMA** https://www.mynrma.com.au/open-road/advice-and-how-to/nrma-transforming-australia-ev-charging-landscape

selectively turn off so as not to have a cluster of chargers overwhelm a transformer at an inopportune moment.

3.7 V2G: Out of Scope but Forward-Compatible

Vehicle-to-Grid discharge is a logical extension of this program and explicitly out of scope at this stage. Bidirectional hardware is more expensive; drivers are wary of expending battery charge cycles without being compensated accordingly; and the regulatory environment is still maturing. These considerations all add extra complexity and risk to an otherwise straightforward proposal.

The OCPP 2.1 hardware specified is forward-compatible with V2G. This initial program builds the operational experience, consumer relationships, and pricing infrastructure a subsequent V2G program would require. A driver who has experienced reliable Solar-Soak charging is a warmer prospect for V2G enrollment than one with no prior grid-interactive charging experience.

As much as the solar soak arrangement depends on chargers located for convenience during the midday peak, any V2G scenario will depend on having reverse-direction chargers located conveniently for grid export when most needed (likely through the early evening). The focus for this equipment would therefore shift from workplaces to homes and evening activity centers.

4. DMIA Pilot Concept

The statewide vision requires a proven, auditable model before it can justify network-wide capital commitment. A DMIA-funded pilot is that first step, and this proposal is based around a nominal 1MW nameplate capacity:

Estimated capital outlay	$750,000 at ~$5,000 per charger installed ● vs 2025-2030 DMIA budget: 10% of $7.18m (Ergon); 12% of $5.93 (Energex) ● vs 2025-2030 Augmentation budget: $432.9m (Ergon); $538.6m (Energex) Demand management software and consumer programming – TBD
Charger count	150 × 7kW OCPP 2.1 AC units Charging hardware to be rolled out ASAP to provide immediate benefit and collect initial utilisation data, without being blocked by software integration.
Aggregate demand shift	1MW nameplate capacity (100% utilisation) during midday solar peak
Regulatory model	Ergon/Energex purchase demand management services from Yurika under a commercial SLA, with Chargefox handling the driver-facing payment layer (builds on existing Yurika-Chargefox relationship).
Charger install locations	Distributed across Ergon/Energex network and selected for: ● Regions with high solar penetration and operational need ● Sufficient regional EV population to support target utilisation rates ● Driver choice and convenience
Outcomes	1. Establish commercial model with Yurika as a first-mover stand-in for future private operators

	2. Demonstrate sub-substation demand management capability in a real regional network 3. Generate utilisation evidence and establish sweet spot <100% 4. Tests the ~$1m per MW thesis
Subsequent steps	Program expansion upwards from 1MW, including via private operators providing demand management capability via AC charging locations.

4.1 Regulatory Pathway

Distribution network service providers cannot use their regulated asset base to fund public EV charging infrastructure, classified as a competitive non-regulated service. Within the Energy Queensland group, this is not a barrier – it points directly to the delivery model.

Yurika as delivery vehicle

Yurika has relevant experience from the Queensland Electric Super Highway project (QESH), including integration with the customer-facing payment platform, Chargefox. This existing integration will be crucial for implementing the “rapid demand management via price signals” component.

The Million Wheel Battery would expand Yurika’s ownership of physical charging infrastructure, carrying the asset on its own balance sheet and recovering costs through charging revenue. The proposed SLA between Ergon and Yurika classifies faulted charging equipment as a grid reliability failure requiring equivalent response urgency to a traditional network fault — reliability is not a customer service consideration, it is load-bearing for the entire business case.

Yurika is the lowest-friction first mover, not the prescribed long-term operator. The pilot's secondary output — beyond grid data — is a proven commercial model that private operators can replicate. Energy Queensland's long-term role is demand management services purchaser, with private operators competing on supply.

DMIA funding scope

The natural fit for DMIA funds is the innovation layer: the demand management software integration, dynamic pricing implementation via Chargefox, OCPP orchestration logic, and consumer behaviour outreach programming to drive high asset utilisation. This is where the genuine novelty lies – the physical chargers are commodity hardware; the demand management capability built on top of them is not.

The proposed hardware provides immediate demand management outcomes the moment it is commissioned. By placing chargers intentionally at solar-dense substations and feeders, the network gains a distributed “passive” solar soak that manages local voltage rise even before the orchestration layer is fully operational.

This immediate utility informs the funding split: the base hardware can be justified as a network capital investment for immediate augmentation deferral, leaving DMIA funds to focus on the "over-and-above" innovation of real-time orchestration. The final determination of who pays for hardware (Yurika self-funding as a commercial investment, a separate Energy Queensland capital program, or private co-investment) will depend on Yurika’s internal capex appetite and Ergon’s non-network option processes.

Victorian AER ring-fencing waiver

Alternatively, a supporting precedent exists in the October 2025 AER ring-fencing waiver granted to Victorian distributors (CitiPower, Powercor, United Energy)[23], which recognised that distributors require direct operational control to test how EV charging manages local grid constraints. This is regulatory validation of EV charging as a network support function rather than a purely competitive service. It's relevant if Ergon were to pursue direct ownership as an alternative to the Yurika model, and may be attractive if operational urgency or scale requires immediate, direct control.

23 https://www.aer.gov.au/industry/networks/ring-fencing/citipower-powercor-and-united-energy-ring-fencing-waiver-electric-vehicle-charging-infrastructure/decision